

FINAL PROJECT README.file

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Location of .SOF file: [CPEN_FINALV2 / rt1 / LAB2V1 / template_de1soc / simple_ipod_solution.sof](#)

All important .v and .psm files are in the **red** directory, above.

Status of the project: project is operational except that

- **Results are shown on the HEX display instead of speech synthesizer**
- **When you press something wrong calculator remains unaffected, but it won't say "error"**

How to run the project:

After uploading the .SOF file into the DE1, try pressing one number, then either plus or minus and after that another number. First number, second number and the operation are said by the speech synthesizer. After you press the second number you should see the results on the HEX display. To make the circuit reset, **YOU HAVE TO PRESS "="**. After pressing "=" you can enter another set of numbers and operations and see the results, verify this if you hear numbers, plus or minus. **RESULTS** are shown in hexadecimal 2's complement numbers.

SIMULATIONS: [CPEN_FINALV2 / sim / FINALPROJSIMV2](#)

simulations are mainly done using ModelSim but I used the HEX display and LEDs to debug my circuit each time I compiled my Verilog code. Look at lines 630 to 636 in [simple_ipod_solution.v](#). ADDRESS_CNTRL which is in between the flash and the PicoBlaze is mainly debugged using those signals in line 630 to 636.

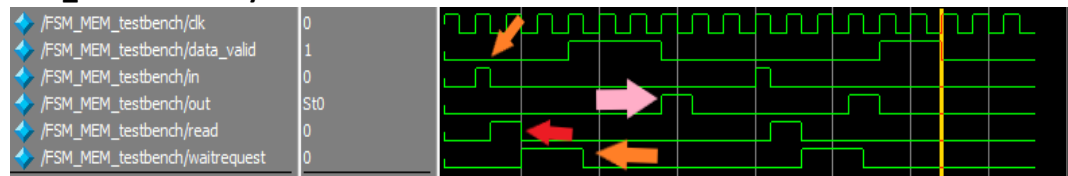
All important simulations are in the **blue** directory.

1. LED averaging:



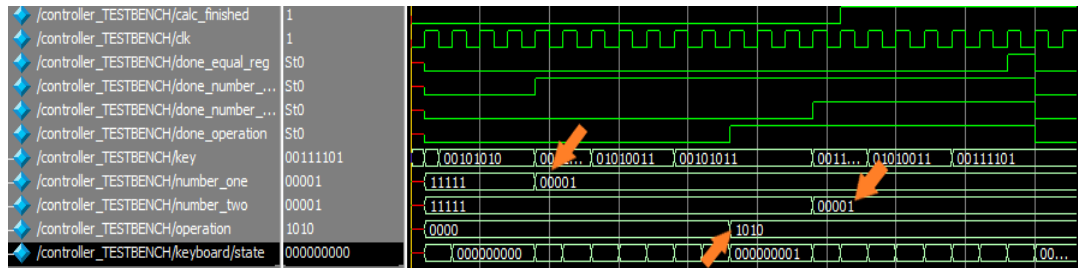
I just wanted to see the count value go up, since it was the only tricky part of this FSM. As you can see on the 5th line count goes up by 1 so this module is expected to be operational.

2. FSM_MEM: memory verifier module



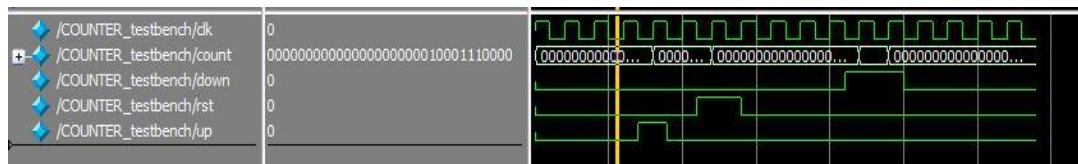
After read, wait_request and data_valid we see the finish out signal, so this module is expected to be operational.

3. KEYBOARD:

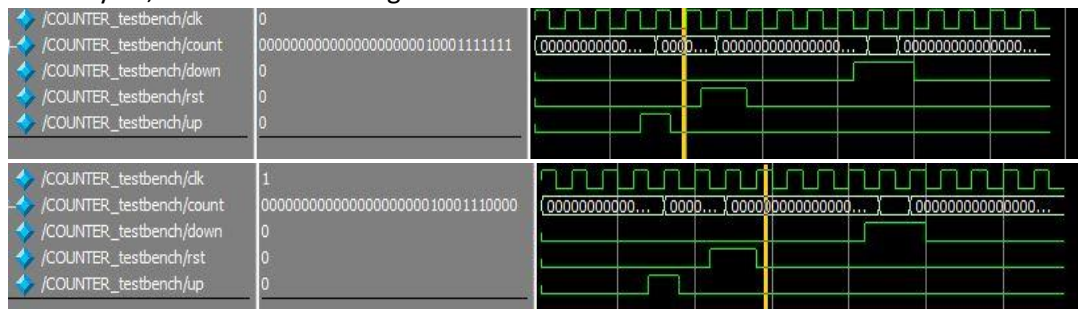


When keys are pressed, values of number_one, number_two and operation, change from the default value to the desired values.

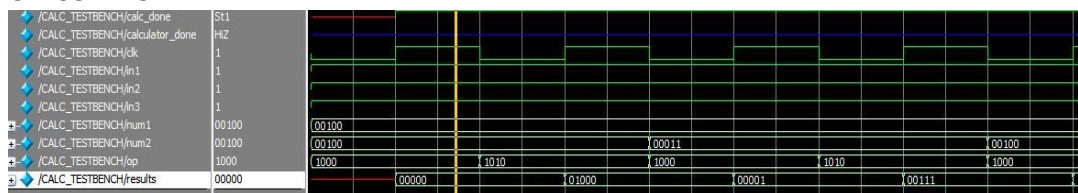
4. COUNTER:



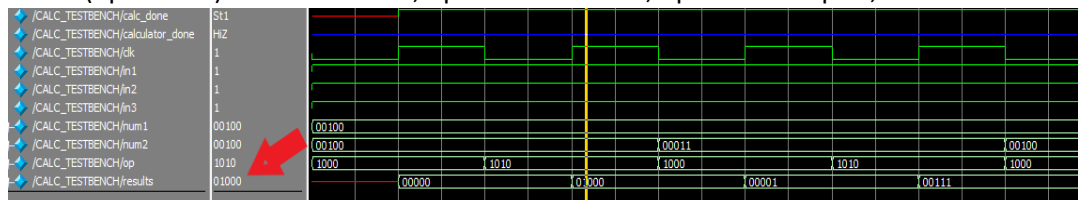
Count is at 1136 (decimal) in the beginning, if up is set it goes up by 15 and if down is set it goes down by 15, reset makes count go to 1136.



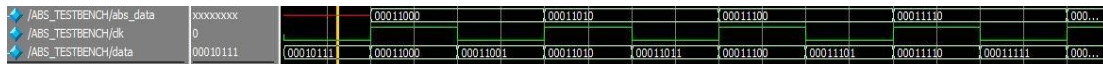
5. CALCULATOR:



num1 (op = + or -) num2 = results ; op = 8 is for minus, op = 10 is for plus;



6. ABS data:



Calculates absolute values before sending audio_data into the LED averaging FSM.

7. CLOCK DIVIDER



KEYS ARE USED to control the speed of the speech, same up and down signals from lab 3 were used. Value of the “speedometer” should be d90 or 1136 in decimal. To test the keys uncomment line 633 and make line 632 commented in **simple_ipod_solution.v**. You could also listen carefully to the audio.

Narrator_cntrl FSM is significantly changed so make sure if you recompile my project in Quartus you use the correct version which is at the bottom of **simple_ipod_solution.v**.

ALL of the synthesizable code written for this project is in three different files as follows:

- **pracPICO.psm**
- **simple_ipod_solution.v**
- **picoblaze_template.v**
- **system.v**

THANK YOU FOR YOUR TIME